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Regularized SPH model for soil–structure interaction with generalized frictional boundary

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Abstract

This paper proposes a regularized smoothed particle hydrodynamics (SPH) scheme to solve the problems of tensile instability and stress noise in large-deformation soil–structure interaction. Particle shifting and stress regularization are combined to maintain uniform stresses and particle distributions. A generalized frictional boundary condition is presented to model soil–structure interfaces with abitrary friction coefficients. A modified particle shifting technique is developed to regularize particles near the frictional boundaries. Furthermore, GPU acceleration algorithm is adopted to improve the efficiency. The proposed scheme is compared with the conventional SPH methods using numerical examples, including the collapse of three-dimensional granular column, the collapse of the cohesive soil, and pipe penetrates into soft clay. The comparison indicates that the proposed regularized SPH model can give highly accurate results for large-deformation soil–structure interaction problems with arbitrary friction coefficient.

KEYWORDS

frictional contact, large deformation, particle regularization, smoothed particle hydrodynamics, soil structure interaction

1 | INTRODUCTION

Large deformation soil–structure interaction can be found in various geotechnical and geological problems, for example, cone penetration test and pile installation,[1,2] seabed–structure interaction,[3,4] and impact of mass movements on structures.[5,6] The large deformation and moving soil–structure interfaces are inconvenient to model in conventional grid-based methods. As a result, alternative methods, based on either constant remeshing[7–9] or meshless approaches,[10–12] are developed to model this type of problem.

Smoothed particle hydrodynamics (SPH) is a truly meshless method becoming popular in recent years because it can easily handle large deformation, moving boundaries, and material interfaces. It was successfully applied to some challenging geotechnical problems, for example, large deformation,[10,11,13] granular flows,[14–17] shear band,[18,19] and dynamic soil–fluid coupling.[20–22] The above applications show that SPH is a capable method. Nevertheless, SPH has some numerical issues that call for further improvements. Two of the issues are tensile instability and noisy stress results.[23]

Previous studies demonstrate that tensile instability in SPH is related to stress states and the second derivative of kernel function.[24–26] When modeling noncohesive geomaterials, tensile instability can be prevented by the tension cracking treatment proposed by Bui et al.[10] For cohesive soils, tensile instability is more critical. The most widely used methods to mitigate it is the artificial stress method,[10,27] which introduces an additional repulsive force to prevent particles from getting too close. Although it can successfully alleviate tensile instability, the artificial stress method has two difficulties: (1) it has one numerical parameter that needs to be tuned to achieve the optimal stabilization effect; (2) it gives rise to noisy stress results, which is usually overlooked in previous studies. Several other attempts have been made to cure tensile instability.[28–30] As one of the representative approaches, the total-Lagrangian SPH (TLSPH) method proves to be capable in effectively eliminating the tensile instability.[31–33] Nevertheless, the TLSPH method is based on the initial configuration, indicating that it is inconvenient for problems involving large deformation because in those cases, distorted deformations may lead to unphysical Jacobians hence breakdowns of computations.[33] It still requires further investigations to develop a general tensile instability treatment method.

Moreover, because most of previous SPH simulations focus on kinematics, the accuracy of stress results attract less attention. However, in soil—structure interaction problems, the accurate prediction of reaction force is crucial. Therefore, it is important to obtain accurate stress results in soils. It is well-known that the classical SPH method gives rise to noisy stress results due to the so-called short-length-scale-noise[14,34] and hourglass mode.[35,36] A simple treatment is to use the Shepard filter to periodically smooth the stress field.[37] However, the Shepard filter has only C^0 consistency and may induce excessive damping and unphysically large stress at surface particles. Nguyen et al.[34] proposed to use a stress regularization technique based on C^1 consistent moving least square (MLS) smoothing, where smooth stress results were obtained. Recently, following the δ-SPH method,[38] Feng et al.[39] introduced a diffusive term in stress integration to smooth out stress fluctuations.

To accurately model large-deformation soil–structure interaction, we need to have a stable and accurate SPH method and an efficient approach for the soil–structure frictional contact. To this end, first we develop a regularized SPH model free of tensile instability and stress noise. This is achieved by two regularization techniques, that is, particle regularization and stress regularization. Based on the observation that tensile instability leads to particle clumps, we employ the particle shifting technique (PST)[40–42] to maintain uniform regularized particle distributions; thus, clumps and unphysical numerical crackings can be avoided. Compared to the tension cracking treatment and artificial stress, PST can be used to mitigate the tensile instability in both noncohesive and cohesive soils without tuning parameters. Furthermore, it gives rise to uniform particle distributions, which are helpful in improving the SPH approximation accuracy. Nevertheless, we find that although PST can give better stress field, it cannot completely eliminate stress noise. Consequently, we propose a stress regularization technique based on the finite particle renormalization,[43] which has also C^1 consistency but less computational cost.[34]

The regularized SPH model is without tensile instability and stress noise in large deformation modeling of geomaterials. However, the reliable simulation of soil–structure interaction also requires accurate modeling of frictional boundaries. In SPH simulations of geotechnical problems, the boundaries are usually modeled using the boundary particle method,[10,14] which can reproduce nonslip and free-slip boundary conditions. The boundary particle method is unable to consider different friction angle; thus, its application is limited for complex soil–structure interaction problems. Several friction contact methods allowing arbitrary wall friction angle were proposed for SPH simulations in the past.[44–48] Among them, the hybrid contact method[47,49,50] combines the convenience of point-to-point contact and the accuracy of point-to-segment contact, and is suitable for problems with arbitrary geometry. In this work, it is combined with the regularized SPH model to solve large deformation soil–structure problems.

The remaining part of the paper is organized as follows. The governing equations and formulations of classical SPH are briefly introduced in Section 2 and 3, respectively. In Section 4, the boundary particle method and the hybrid contact method for frictional soil–structure interaction are detailed. The regularization techniques for particle position and stress are presented in Section 5. Numerical examples are presented in Section 6, and some conclusions are given in Section 7.

2 | GOVERNING EQUATIONS

2.1 | Conservation equations

For large deformation soil–structure interaction problems, the governing equations for soils can be written in the following Lagrangian form[45]:

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -\rho\nabla\cdot\mathbf{v} \tag{1}$$

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{1}{\rho}\nabla\cdot\boldsymbol{\sigma} + \mathbf{g} + \mathbf{a}^e, \tag{2}$$

where ρ is the soil density, $\mathbf{v}$ denotes the velocity, $\boldsymbol{\sigma} = \mathbf{s} - p\mathbf{I}$ is the Cauchy stress tensor where $\mathbf{s}$ is the deviatoric stress and $p = -(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})/3$ is the pressure, $\mathbf{I}$ is a 3×3 identity matrix, the subscripts x, y, and z are the directions of the spatial coordinates; $\mathbf{g}$ and $\mathbf{a}^e$ stand for the acceleration vectors caused by gravity and other external force, respectively. In this study, the external force mainly comes from soil–structure interaction. $\mathrm{d}(\cdot)/\mathrm{d}t$ represents the material time derivation and ∇ is the gradient operator.

2.2 | Constitutive equations

To close Equations (1) and (2), an elastic–perfectly plastic model is employed to describe soil behavior. The Drucker–Prager (DP) yield criterion and a nonassociated flow rule are adopted, which read

$$f = \sqrt{J_2} - \alpha_\phi p - \alpha_c c \quad \text{and} \quad g = \sqrt{J_2} - \alpha_\psi p, \tag{3}$$

where $J_2 = \mathbf{s} : \mathbf{s}/2$ is the second invariants of the stress tensor, α_ϕ and α_c are the constitutive parameters related to the friction angle ϕ and cohesion c from Mohr–Coulomb model

$$\alpha_\phi = \frac{6\sin\phi}{\sqrt{3}(3-\sin\phi)} \quad \text{and} \quad \alpha_c = \frac{6\cos\phi}{\sqrt{3}(3-\sin\phi)}, \tag{4}$$

where the parameter α_ψ is related to the dilatancy angle ψ in a similar way as α_ψ is related to ψ. The details of the implementation of the DP model in SPH computations can be found in the pioneering work of Bui et al.[10]

3 | CLASSICAL SPH FOR GEOMATERIALS

In this section, we briefly introduce the formulations and computational procedures of the conventional SPH method employed in geotechnical simulations. For more details, interested readers can refer to Bui et al.[10] and Peng et al.[37]

3.1 | Framework of SPH

In SPH, the computational domain can be discretized by a set of particles representing material elements. The value of each field function $f(\mathbf{x})$ at a location $\mathbf{x}$ can be expressed using the following equation over the integral domain Ω[51]:

$$f(\mathbf{x}) = \int_\Omega f(\mathbf{x}')W(\mathbf{x}-\mathbf{x}', h)\mathrm{d}\mathbf{x}' \tag{5}$$

in which $f(\mathbf{x}')$ denotes the surrounding field variable at $\mathbf{x}'$ within the integral domain Ω; W is the so-called kernel function and h denotes the smoothing length defining the size of domain Ω. In SPH, a field function $f(\mathbf{x})$ at a particle can be approximated through the weighted summation of the function values at its neighboring particles as follows:

$$\langle f(\mathbf{x}_i)\rangle \approx \sum_j \frac{m_j}{\rho_j} f(\mathbf{x}_j) W(\mathbf{x}_i - \mathbf{x}_j, h) \tag{6}$$

$$\langle \nabla f(\mathbf{x}_i)\rangle \approx \sum_j \frac{m_j}{\rho_j} f(\mathbf{x}_j) \nabla_i W(\mathbf{x}_i - \mathbf{x}_j, h), \tag{7}$$

where the bracket $\langle\rangle$ indicates the approximation value; the subscripts i and j denote the concerned particle i and its neighboring particle j, respectively; m is the mass; the symbol $\sum$ represents the summation of neighboring particles for the concerned particle, and ∇_i represents the gradient evaluated at $\mathbf{x}_i$. More details about Equations (6) and (7) can be found in the literature.[51] In this work, the Wendland C2 kernel[37] is adopted because of its good performance in preventing particle clumping.[52] h is set to $1.5\Delta x$ where Δx is the initial particle spacing.

With the SPH discretization, the continuous governing equations (1) and (2) are usually written as the following forms in geotechnical simulations:

$$\frac{\mathrm{d}\rho_i}{\mathrm{d}t} = \sum_j m_j (\mathbf{v}_i - \mathbf{v}_j) \cdot \nabla_i W_{ij} \tag{8}$$

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = \sum_j m_j \left(\frac{\boldsymbol{\sigma}_i}{\rho_i^2} + \frac{\boldsymbol{\sigma}_j}{\rho_j^2} + \Pi_{ij}\mathbf{I} \right) \cdot \nabla_i W_{ij} + \mathbf{g}_i + \mathbf{a}_i^e, \tag{9}$$

in which $W_{ij} = W(\mathbf{x}_i - \mathbf{x}_j, h)$. The dissipative term Π_{ij} denotes the artificial viscosity used to prevent unphysical oscillations. Its detailed formulation can be found in Refs. 10 and 53.

The above governing equations are ordinary partial differential equations, which are solved using explicit time integration schemes. In this study, a leap-frog integration scheme is employed with CFL condition controlled time step.[10] Moreover, while the frictional boundary method is employed, the additional constraint from acceleration controlled time step is needed.[50]

3.2 | Stress update in SPH

For large deformation problems, the stress can be obtained by integrating the following Cauchy stress rate:

$$\dot{\boldsymbol{\sigma}} = \boldsymbol{\sigma} - \boldsymbol{\sigma} \cdot \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \boldsymbol{\sigma}, \tag{10}$$

where $\boldsymbol{\sigma}$ denotes Jaumann stress rate used to keep the frame objectivity in large deformation simulations. $\boldsymbol{\omega} = (\mathbf{L} - \mathbf{L}^{\mathrm{T}})/2$ is the spin tensor, where $\mathbf{L} = \nabla \otimes \mathbf{v}$ denotes the velocity gradient that can be obtained using the following SPH approximation

$$\mathbf{L}_i = \sum_j \frac{m_j}{\rho_j} (\mathbf{v}_j - \mathbf{v}_i) \otimes \nabla_i W_{ij}. \tag{11}$$

Based on the elastoplastic constitutive model, the trial stress for a purely elastic response σ^* can be written as follows:

$$\boldsymbol{\sigma}^* = \boldsymbol{\sigma}^t + (2G\dot{\mathbf{e}} + K\dot{\varepsilon}_v \mathbf{I} - \sigma^t \cdot \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \sigma^t)\Delta t, \tag{12}$$

where $\dot{\mathbf{e}} = \dot{\boldsymbol{\varepsilon}} - (\dot{\varepsilon_v}/3)\mathbf{I}$ is the deviatoric strain rate tensor, in which $\dot{\boldsymbol{\varepsilon}}$ is defined as $\dot{\boldsymbol{\varepsilon}} = (\mathbf{L} + \mathbf{L}^{\mathrm{T}})/2$, the volumetric strain rate is $\dot{\varepsilon_v} = \dot{\varepsilon}_{xx} + \dot{\varepsilon}_{yy} + \dot{\varepsilon}_{zz}$. G and K are the shear and bulk modulus, respectively. The superscript t is the time at the beginning of the computational step, and Δt is the size of time step. The way to update the stress tensor can be summarized as the following three steps[54]:

(1) Yielding deformation occurs if $f(\sigma^*) > 0$;
(2) The incremental plastic multiplier $\Delta\lambda$ can be derived through the Taylor-series expansion of the yield function;
(3) The return mapping algorithm with backward Euler scheme is adopted to solve the problems of tension cracking.[10]

4 | SPH MODELING OF SOIL–STRUCTURE INTERACTION

In SPH simulations, structures can either be considered as rigid[47,55] or deformable.[32,44] To model soil–structure interaction, it is crucial to consider the actual boundary condition along the interface, as well as accurately capture the interaction forces. Most of previous studies employ the boundary particle method,[10,37] which achieves the nonslip boundary condition originated from fluid mechanics. For this method, the effective friction angle at the boundary interface can be obtained, and the actual friction effect of the boundary particle method has been quantitatively assessed in del Castillo et al.[56] Moreover, to simulate general soil–interaction problems, numerical schemes that consider the frictional contact are needed. In this study, two methods, that is, the boundary particle method and the hybrid contact method are employed to model soil–structure interaction. Their concepts are introduced in this section, while performance comparisons of these two methods will be investigated with numerical examples subsequently.

4.1 | Boundary particle method

In the boundary particle method, boundaries are discretized using dummy particles. The boundaries are considered to have the same material properties as soil but remain undeformed with prescribed motions. This can effectively realize a nonslip boundary condition, which was believed to reproduce rough contact with a contact friction angle equal to 0.79 times that of the soil.[56]

In this method, the stresses at boundary particles are extrapolated from soil particles using the following formulation[37,57]:

$$\sigma_b^{lk} = \frac{\sum_i \sigma_i^{lk} W_{bi}}{\sum_i W_{bi}}, \quad l \neq k; \quad \sigma_b^{ll} = \frac{\sum_i \sigma_i^{ll} W_{bi} - g_b^l \sum_i \rho_m r_{bi}^l W_{bi}}{\sum_i W_{bi}} \tag{13}$$

where the subscripts b and i denote boundary and soil particles, respectively; the superscripts $l, k = x, y, z$ are the directions of the spatial coordinates, and Einstein convention does not apply; r means the distance vector from particle i to b. The completed momentum conservation equation of the material particles can be obtained by substituting Equation (13) into Equation (9). The boundary particles take part in the computation of the momentum equation like soil particles. The only difference is that the field variables at boundary particles are either prescribed (density and velocity) or extrapolated (stress), not calculated. The total contact force can be obtained by summing up all the interaction forces of the boundary particles. It can be observed that in the boundary particle method, the boundary friction is not explicitly considered. By assuming that the boundary has the same property as the soil, it approximately mimics a perfectly rough contact condition.

4.2 | Generalized frictional boundary

With the aim to directly model the soil-boundary contact, several particle-to-particle and particle-to-segment methods were proposed for SPH.[44,45,58,59] In general, the particle-to-particle methods are simple and easy to implement, but less accurate.[58] On the other hand, the particle-to-segment methods are more accurate,[44,47] but may have high cost and are not straightforward to implement due to the need of contact detection and the treatment of many special cases.[60] Besides, as SPH is a truly meshless method, the particle-to-segment methods needs additional efforts in generating surface segments (in 2D) and patches (in 3D).[61]

The hybrid contact method[47,49,62] provides an alternative for the contact analysis in SPH. It has similar accuracy as the particle-to-segment methods but is as convenient as the particle-to-particle methods. It also has the advantage of being

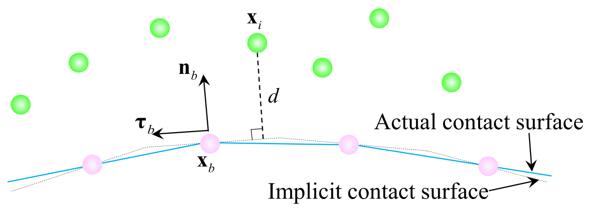

FIGURE 1 The schematic diagram of contact detection.[62]

applicable to problems with complex geometry. The concept of the hybrid contact method is shown in Figure 1. In the hybrid contact method, the structure surface is discretized using one layer of structure nodes, no surface mesh is needed. Note that the discretization of the structure surface is independent from the particle spacing of the soil phase, which gives more flexibility in numerical simulations. At each of the structure node, the local surface normal is obtained from the geometry information. The contact calculation has the following four steps.

(1) For a soil particle $\mathbf{x}_i$, find its nearest structure node $\mathbf{x}_b$. The distance between the soil–structure particle-node pair in the direction of the local surface normal is computed as

$$d = (\mathbf{x}_i - \mathbf{x}_b) \cdot \mathbf{n}_b \tag{14}$$

where $\mathbf{n}_b$ is the unit normal vector at the structure node $\mathbf{x}_b$.

(2) The contact is activated if

$$d < \beta \Delta x \tag{15}$$

where $\beta = 0.5 - 1$ is a coefficient that controls the position of contact surface. For instance, in common SPH discretization, as the particle represents a material element, the actual surface usually is $0.5\Delta x$ away from the particle location. In this case, β should be 1.0. On the other hand, if the structure node locates exactly on the structure surface, β should be 0.5.

(3) After the contact detection, the normal contact force $\mathbf{f}^n$ can be calculated based on the criteria of partial penetration[50]

$$\mathbf{f}^n = (k^n d_p - \eta^n \Delta \mathbf{v}^n)\mathbf{n}_b \tag{16}$$

in which $k^n = \xi m_i/(\Delta t)^2$ and ξ is a coefficient that controls the extent of how much penetration is allowed; $d_p = \beta\Delta x - d$ is the penetration; $\eta^n = 2\ln(e_n)\sqrt{k^n m_i}/\sqrt{\ln^2(e_n) + \pi^2}$ denotes the damping coefficient in the normal direction where e_n represents the constant of restitution. The relative velocity in the normal direction $\Delta\mathbf{v}^n$ is written as

$$\Delta\mathbf{v}^n = [\mathbf{n}_b \cdot (\mathbf{v}_i - \mathbf{v}_b)]\mathbf{n}_b \tag{17}$$

where $\mathbf{v}_i$ and $\mathbf{v}_b$ are the velocity of the soil and structure nodes, respectively. Note that the velocity of the structure node $\mathbf{v}_b$ is prescribed as boundary condition in this study. The problem of soil–structure interaction involving deformable structures is not considered here. The calculation for rigid structures with translation and rotational displacement can be found in Refs. 45 and 50.

(4) To account for the effects of friction and cohesion, the tangential contact force $\mathbf{f}^\tau$ is given as

$$\mathbf{f}^\tau = \begin{cases} -(\mu\|\mathbf{f}^n\| + \mu_c c A_p)\boldsymbol{\tau}_b, & \|\mathbf{f}_s^\tau\| \gg \mu\|\mathbf{f}^n\| + \mu_c c A_p \\ \mathbf{f}_s^\tau, & \|\mathbf{f}_s^\tau\| < \mu\|\mathbf{f}^n\| + \mu_c c A_p \end{cases} \tag{18}$$

where $\mathbf{f}_s^\tau = -k^\tau\|\boldsymbol{\delta}^\tau\|\boldsymbol{\tau}_b$ is the tangential contact force in static contact status, $k^\tau = 2k^n/7$ is the tangential stiffness,[63] $\boldsymbol{\delta}^\tau = \int \Delta\mathbf{v}^\tau dt$ denotes the accumulated relative displacement in the tangential direction, $\Delta\mathbf{v}^\tau = (\mathbf{v}_i - \mathbf{v}_b) - \Delta\mathbf{v}^n$ is the relative tangential velocity between the soil particle and structure node, $\boldsymbol{\tau}_b = \Delta\mathbf{v}^\tau/\|\Delta\mathbf{v}^\tau\|$ represents the

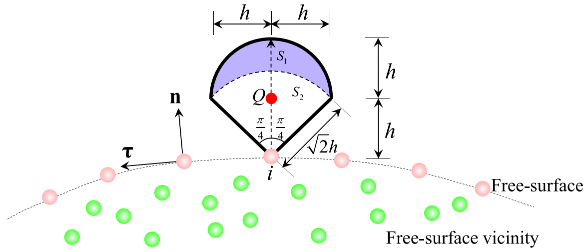

FIGURE 2 Subgroups of particles in PST with the boundary particle method.

tangential direction of the contact, $\mu = \tan\phi$ is the frictional coefficient, μ_c denotes the reduction coefficient related to soil cohesion c, and A_p is the contact area for one particle, which is Δx in 2D and Δx^2 in 3D.

(5) Finally, for the soil particle i, the acceleration related to the contact force is obtained as

$$\mathbf{a}_i^e = \frac{\mathbf{f}^n + \mathbf{f}^\tau}{m_i} \tag{19}$$

The completed momentum conservation equation of the soil particles can be obtained by substituting Equation (19) into Equation (9). More details related to $\mathbf{f}^n$ and $\mathbf{f}^\tau$ can be found in Refs. 32, 50. Note that if the free-slip (frictionless) boundary is employed, Equation (18) is equal to zero.

5 | REGULARIZATION TECHNIQUES

When the classical SPH method (Section 3) is used to model soil–structure interaction, the simulations usually suffer from tensile instability and low stress accuracy. To avoid tensile instability and achieve accurate stress results at the same time, we introduce two regularization techniques. The first one is particle shifting to prevent tensile instability by maintain uniform particle distribution, the second is the first-order stress regularization using the finite particle renormalization.

5.1 | Particle shifting technology

The particle shifting technology (PST) is becoming more popular in SPH to ensure uniform particle distribution and improve accuracy.[40,64–66] However, its effectiveness in mitigating tensile instability in solid computations has not been investigated yet. In this part, the PST and how it can be adjusted to consider the friction boundary are presented.

5.1.1 | Free-surface detection

In PST, special treatments should be adopted for the free-surface and the neighboring particles. As explained in Lind et al.,[41] numerical instabilities can develop on the free surface as a result of incomplete kernel support and increased kernel interpolation error. Therefore, it is first crucial to detect surface particles.[41] Usually, the following two-step approach is employed[67]: (1) perform a rough surface detection using the position divergence or eigenvalues of the renormalization matrix; (2) carry out a geometrical scan to select the actual free-surface particles. In this work, because we employ GPU parallelization, it is inconvenient to have a separate subgroup of potential free-surface particles and then use geometrical scan to identify the true ones. Therefore, we only employ the geometrical scan to detect the free-surface particles, which simplifies the implementation and reduces the computational cost.

The "umbrella-shaped" geometrical scan method,[67] as shown in Figure 2, is adopted in this study. The direction of the umbrella area is co-axial with the local unit normal vector $\mathbf{n}_i$, which is defined as [64]

$$\mathbf{n}_i = -\frac{\mathbf{L}_i \nabla \mathbf{C}_i}{\|\mathbf{L}_i \nabla \mathbf{C}_i\|} \tag{20}$$

where ∇C_i and $\mathbf{L}_i$ denote, respectively, the concentration gradient and renormalization matrix of particle i with the following definitions[68]:

$$\nabla C_i = \sum_j \frac{m_j}{\rho_j} \nabla_i W_{ij} \quad \text{and} \quad \mathbf{L}_i = \left[-\sum_j \frac{m_j}{\rho_j} (\mathbf{x}_i - \mathbf{x}_j) \otimes \nabla_i W_{ij} \right]^{-1} \tag{21}$$

For Equation (20), there is no difference between nonslip boundary and contact boundary schemes. Note here all the SPH particles, that is, soil particles and boundary particles, contribute to the computations of ∇C_i and $\mathbf{L}_i$, whereas the structure nodes are not considered as they do not participate in the regular SPH kernel approximations.

With the local unit normal vector, two subareas can be defined, namely S_1 and S_2, as shown in Figure 2. If there is a particle j, which locates in area S_1 or S_2, the following conditions are satisfied[67]:

$$\begin{cases} \forall j \in \left[\|\mathbf{x}_{ij}\| \geq \sqrt{2}h,\ \|\mathbf{x}_{Qj}\| < h \right] & \Rightarrow j \in S_1 \\ \forall j \in \left[\|\mathbf{x}_{ij}\| < \sqrt{2}h,\ \left(\|\mathbf{n}_i \cdot \mathbf{x}_{Qj}\| + \|\boldsymbol{\tau}_i \cdot \mathbf{x}_{Qj}\| \right) < h \right] & \Rightarrow j \in S_2 \end{cases} \tag{22}$$

in which Q is a point that has a distance h from i in the direction of $\mathbf{n}_i$, that is, $\mathbf{x}_Q = \mathbf{x}_i + h\mathbf{n}$. $\boldsymbol{\tau}_i$ is the unit tangential vector, computed as $\boldsymbol{\tau}_i = (-n_{iy}, n_{ix})^{\mathrm{T}}$, where n_{ix} and n_{iy} are the components of $\mathbf{n}_i$. Note that Equation (22) is only applicable for two-dimensional problems.

Consequently, the surface particles can be regarded as those ones with no neighboring particles in the "umbrella-shaped" area [67]

$$\begin{cases} \forall j \notin S_1 \cup S_2 & \Rightarrow i \in \mathbb{F} \\ \text{Otherwise} & \Rightarrow i \notin \mathbb{F} \end{cases} \tag{23}$$

where $\mathbb{F}$ denotes subgroup of free-surface particles. In the GPU-based implementation, Equation (23) is checked for each soil particle and all free-surface particles are identified in one step.

5.1.2 | PST with the boundary particle method

At the end of each step, the particle positions are shifted slightly based on the local particle distribution. Based on the Fick's law, the shifting vector is defined as[41]

$$\delta \mathbf{x}_i = -2h\Delta t \|\mathbf{v}\|_{\max} \nabla \bar{C}_i \tag{24}$$

where $\|\mathbf{v}\|_{\max}$ is the maximum velocity magnitude of the soil particles, $\nabla \bar{C}_i$ is the modified concentration gradient, whose definition is different for different particle subgroups. In SPH simulations with the boundary particle method, the SPH particles can be categorized into four subgroups: the boundary particles $\mathbb{B}$, the free-surface particles $\mathbb{F}$, the free-surface vicinity particles $\mathbb{V}$, and the internal particles $\mathbb{I}$, as shown in Figure 3. The free-surface vicinity particles are identified as those soil particles that have distance less than kh from the free-surface. The distance l_v between a particle and the free-surface can be computed as the distance between a particle and the nearest free-surface particle. All soil particles do not belong to $\mathbb{B}$, $\mathbb{F}$, and $\mathbb{V}$, which are grouped as internal particles.

Following Wang et al.,[65] in this study, the PST is only employed for particles from $\mathbb{V}$ and $\mathbb{I}$. For particles from the two subgroups, the modified concentration gradient is written as

$$\nabla \bar{C}_i = \begin{cases} \displaystyle\sum_j \frac{m_j}{\rho_j} f_{ij} \nabla_i W_{ij}, & \text{if } \|\mathbf{x}_i - \mathbf{x}_j\| < l_v \text{ and } \mathrm{i} \in \mathbb{V} \\ \displaystyle\sum_j \frac{m_j}{\rho_j} (1 + f_{ij}) \nabla_i W_{ij}, & \text{if } \|\mathbf{x}_i - \mathbf{x}_j\| < 2h \text{ and } \mathrm{i} \in \mathbb{I} \end{cases} \tag{25}$$

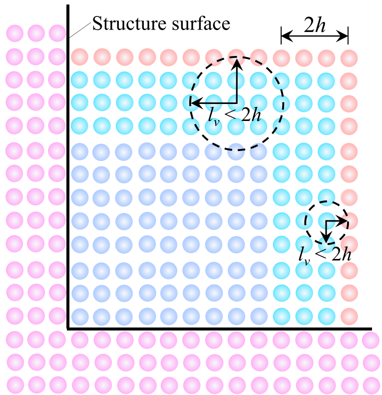

FIGURE 3 Subgroups of particles in PST with the boundary particle method.

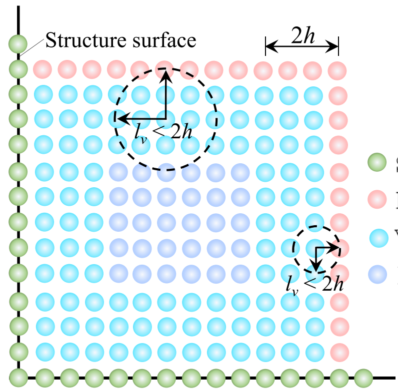

FIGURE 4 Subgroups of particles in PST with the frictional boundary.

where f_{ij} originally comes from the artificial stress method[69]

$$f_{ij} = 0.2\left(\frac{W_{ij}}{W_{ij}(\Delta \mathbf{x})}\right)^n \tag{26}$$

in which $n = W_{ij}(0)/W_{ij}(\Delta x)$. Note that when computing $\nabla \bar{C}_i$ at the surface vicinity particles, the neighboring particles with distance $l_v \leq \|\mathbf{x}_i - \mathbf{x}_j\| \leq 2h$ are excluded to avoid the deficiency caused by the free-surface.

5.1.3 | PST with the generalized boundary

In SPH simulations with the frictional boundary, the shifting vector at each soil particle is also computed using Equation (24). However, when using the frictional boundary, there is no boundary particles on the structure. Consequently, the particles near the structure surface cannot be identified as internal particles. Instead, they are grouped as vicinity particles, as shown in Figure 4. Same as the approach used in the PST with the boundary particle method, first, the free-surface particles are identified using the method introduced in Section 5.1.1. Then, all soil particles having distance less than $2h$ from any free-surface particle or structure node are identified as surface vicinity particles. Finally, all remaining soil particles are grouped as internal particles.

No shifting is applied to free-surface particles. As for the surface vicinity and internal particles, the computation of the modified concentration gradient follows Equation (25). The reduced support domain radius l_v for surface vicinity particles is defined as the smallest distance from the considered soil particle to the free-surface particles and structure nodes. Through numerical experiments, it is found that the particle velocity is sensitive to contact force thus has fluctuations. This velocity fluctuation may give unphysical estimation of the maximum velocity magnitude $\|\mathbf{v}\|_{\max}$, which may result in large shifting vector $\delta\mathbf{x}$ and excessive diffusion. To avoid this, the soil particles in active contact with the structure are excluded when computing the maximum velocity magnitude $\|\mathbf{v}\|_{\max}$.

5.2 | Stress regularization

Shepard filter is the most common method to get better stress results in SPH-based geotechnical simulations.[34,37,47] The original Shepard filter is only C^0 consistent; thus, it may induce nonphysical behaviors in areas with particle deficiency. Nguyen et al.[34] proposed a filter based on MLS interpolation, which has C^1 consistency and performs better than the original Shepard filter. Here, a corrected Shepard stress filter based on the finite particle method[43] is proposed. The corrected Shepard filter has also C^1 consistency and needs only minimal additional computational cost.

For the stress field σ, with Taylor series expansion in multidimensional space can be written as[43]

$$\sigma(\mathbf{x}) = \sigma_i + \nabla\sigma_i \cdot \Delta\mathbf{x} + o(\Delta\mathbf{x}^2) \tag{27}$$

where $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}_i$. Multiplying both sides of Equation (27) with $W(\Delta\mathbf{x})$ or $\nabla W(\Delta\mathbf{x})$, and neglecting the high-order derivatives, we can obtain

$$\int \sigma(\mathbf{x})W(\Delta\mathbf{x})\mathrm{d}\mathbf{x} = \sigma_i \int W(\Delta\mathbf{x})\mathrm{d}\mathbf{x} + \nabla\sigma_i \cdot \int \Delta\mathbf{x}W(\Delta\mathbf{x})\mathrm{d}\mathbf{x} \tag{28}$$

$$\int \sigma(\mathbf{x})\nabla W(\Delta\mathbf{x})\mathrm{d}\mathbf{x} = \sigma_i \int \nabla W(\Delta\mathbf{x})\mathrm{d}\mathbf{x} + \nabla\sigma_i \cdot \int \Delta\mathbf{x} \otimes \nabla W(\Delta\mathbf{x})\mathrm{d}\mathbf{x} \tag{29}$$

The smoothed result of each stress component σ_i^{lk} can be obtained through solving the SPH summation forms of Equations (28) and (29). Thus, with the modified Shepard filter, the smoothed new stress component can be given as

$$\sigma_i^{lk} = \begin{vmatrix} \sum_j \frac{m_j}{\rho_j}\sigma_j^{lk} W_{ij} & \sum_j \frac{m_j}{\rho_j}\mathbf{x}_{ji}^{\mathrm{T}} W_{ij} \\ \sum_j \frac{m_j}{\rho_j}\sigma_j^{lk}\nabla_i W_{ij} & \sum_j \frac{m_j}{\rho_j}\mathbf{x}_{ji}^{\mathrm{T}} \otimes \nabla_i W_{ij} \end{vmatrix} \Bigg/ \begin{vmatrix} \sum_j \frac{m_j}{\rho_j} W_{ij} & \sum_j \frac{m_j}{\rho_j}\mathbf{x}_{ji}^{\mathrm{T}} W_{ij} \\ \sum_j \frac{m_j}{\rho_j}\nabla_i W_{ij} & \sum_j \frac{m_j}{\rho_j}\mathbf{x}_{ji}^{\mathrm{T}} \otimes \nabla_i W_{ij} \end{vmatrix} \tag{30}$$

Similar to the original Shepard filter, the modified Shepard filter is also employed every few steps.[34] It should be noted that there are various possibilities to prevent the tensile instability and noisy stress, for example, stress-particle SPH method,[70] or different stress regularization techniques.[71,72] In this paper, only the prediction of SPH scheme with PST is focused on.

6 | NUMERICAL IMPLEMENTATION

In this paper, the second-order accurate leap-frog algorithm is adopted for the time integration[10]

$$\mathbb{E}^{t+\Delta t/2} = \mathbb{E}^{t-\Delta t/2} + \Delta t\mathbb{F}^t \tag{31}$$

$$\mathbf{x}^{t+\Delta t} = \mathbf{x}^t + \Delta t\mathbf{v}^{t+\Delta t/2} + \delta\mathbf{x} \tag{32}$$

where $\mathbb{E}$ denotes a variable (ρ, $\mathbf{v}$, σ or $\boldsymbol{\varepsilon}$) and $\mathbb{F} = \mathrm{d}\mathbb{E}/\mathrm{d}t$ stands for its rate of change; Equation (32) is the final position embedding the shifting vector from Equation (24). Meanwhile, to ensure the stability of the above scheme, the time step size is governed by the Courant–Friedrichs–Levy condition[10,50]

$$\Delta t = 0.1\min\left(\frac{h}{c_0}, \sqrt{\frac{h}{\|\mathbf{a}\|_{\max}}}, \frac{h}{10\|\mathbf{v}\|_{\max}}\right) \tag{33}$$

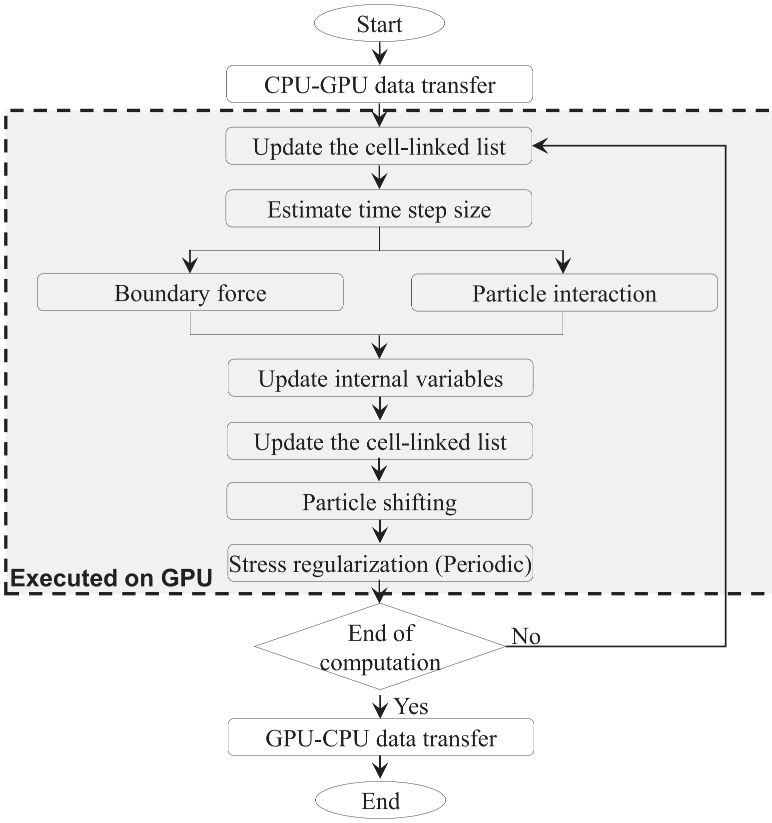

FIGURE 5 The flowchart of computation.

where $c_0 = \sqrt{G/\rho}$ is the sound speed; $\|\mathbf{a}\|_{\max}$ and $\|\mathbf{v}\|_{\max}$ in Equation (33) are the maximum acceleration and velocity magnitude, respectively. Note that $\|\mathbf{a}\|_{\max}$ and $\|\mathbf{v}\|_{\max}$ in Equation (33) are selected from all SPH particles and structure nodes to maintain the stability of the calculation.

The whole computational process is shown in Figure 5. At the beginning of each computational step, the cell-linked list is created for all particles, and the time step size is estimated by the CFL condition. Then the boundary force and interaction force are calculated, respectively. After that, the variables involving position, velocity, Cauchy stress, and the plastic strain can be updated. Next, the cell-linked list is updated. The final position of particles is obtained using PST. The stress regularization is performed periodically at the end of the computational step. All the above schemes are parallelized using GPU. The implementation is based on the open-source solver LOQUAT.[37]

7 | NUMERICAL EXAMPLES

In this section, the regularized SPH model with generalized frictional boundary is tested by comparing numerical results with experimental data or theoretical results. For the modeling of soil–structure interaction, both the boundary particle method and the hybrid contact method are employed. In the boundary particle method, the structures are represented using three layers of dummy particles. In all simulations, the value of β is equal to 0.5; the constant of restitution $e_n = 0.01$.

The proposed SPH scheme is first employed for the gravity-driven granular collapse in three-dimension (3D).[73,74] This case is to show how the boundary friction impacts the flow dynamics, which cannot be modeled using the conventional boundary particle method. As the considered material is cohesionless, there is no tensile instability; thus, the PST method is not used in this case.

For cohesive soil with the softening behavior, the phenomenon of tensile instability and noisy stress can be avoided through the proposed joint scheme of PST combined with stress regularization technology.[10,75] To demonstrate this, two more numerical examples, that is, collapse of cohesive soils and pipe penetration, are included.

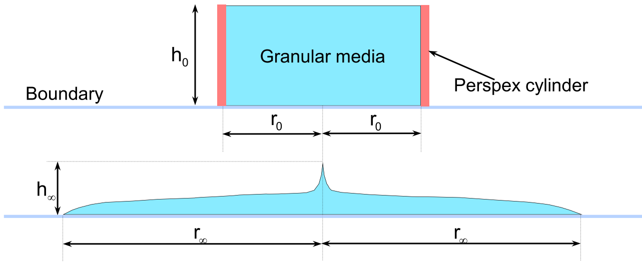

FIGURE 6 The schematic diagram of the 3D granular column.

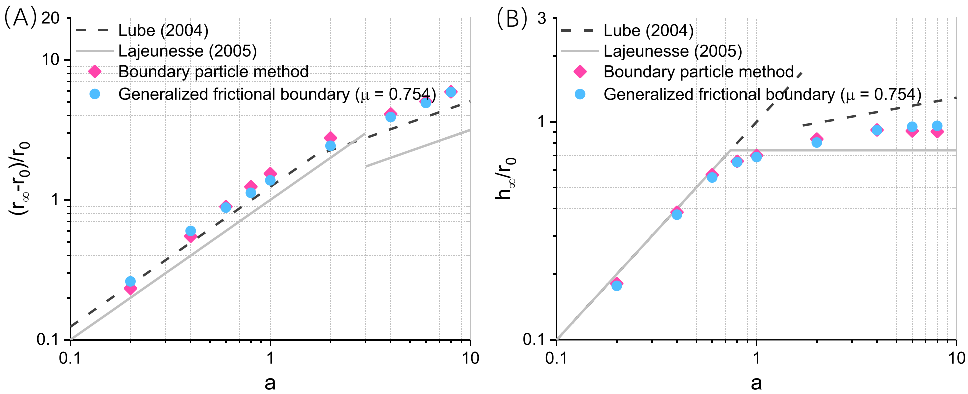

FIGURE 7 Comparison between the simulation and the experiment for 3D granular collapse progress.[73,74] (A) Normalized run-out distance; (B) normalized final height.

7.1 | Collapse of 3D granular column

The problem of collapse of granular column (see Figure 6) was experimentally studied by Lube et al.[73] and Lajeunesse et al.,[74] and later was simulated using various numerical methods. In previous SPH simulations,[15,37,76] the frictional effect between the granular material and the plate is usually considered using the nonslip condition with the boundary particle method except for Sheikh et al.,[46] in which a frictional boundary method is employed.

In the test, a column of granular material is confined in a cylinder. The initial height and radius of the granular body is h_0 and r_0, respectively. During the test, the cylinder is quickly lifted, then the column rapidly collapses on the horizontal surface under gravity and eventually reaches the final deposition. The final height h_∞ and the run-out distance r_∞ of the deposit are dependent on the initial aspect ratio $a = h_0/r_0$.

Previous studies show that the final deposit mode is mainly determined by the aspect ratio a. The shape of the final deposit is slightly affected by the intergranular friction angle ϕ and dilatancy angle ψ,[15,37] and independent of the other material parameters such as elastic modulus E, Poisson's ratio ν and density ρ.[77] In this study, the initial radius r_0 is fixed as 0.2, while the aspect ratio a changes from 0.2 to 10. The initial particle spacing is $\Delta x = 0.01$; the other material parameters can be found in literature[37,73] including: $E = 5$ MPa, $\nu = 0.3$, internal friction angle $\phi = 37°$, cohesion $c = 0$ kPa, $\psi = 0°$, and $\rho = 1850\text{kg/m}^3$. When using the generalized frictional boundary, the following constants are employed: $\xi = 0.01$,[44] $\mu_c = 0$ as cohesionless granular material is modeled.

7.1.1 | Comparison of the two soil–structure interaction methods

First, we investigate the comparison between the boundary particle method and the generalized frictional boundary, as well as the actual effect of the boundary particle method in modeling friction. For both methods, 9 three-dimensional simulations are performed with different aspects ratios. In the simulations with the generalized frictional boundary, the surface friction coefficient is taken as $\mu = \tan\phi$ to model the rough basal surface made of sandpaper. The normalized final run-out distance and height obtained by SPH simulations are shown in Figure 7. The empirical equations proposed by

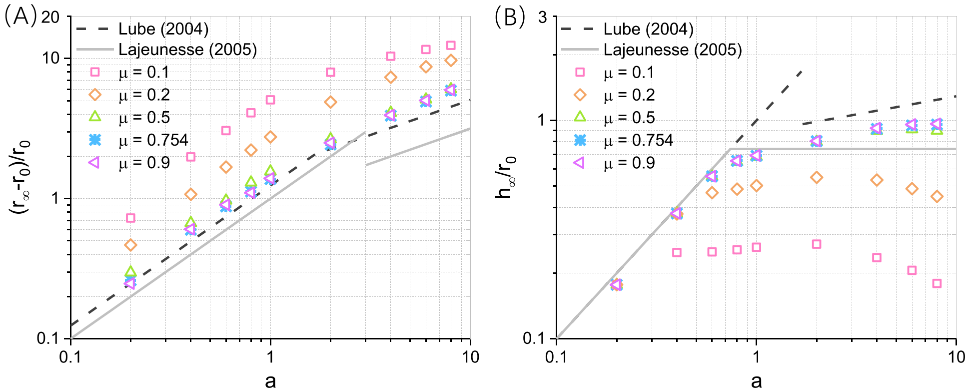

FIGURE 8 Comparison between the simulation and the experiment for 3D granular collapse progress (with varying surface friction μ).[73,74] (A) Normalized run-out distance; (B) normalized final height.

Lube et al.[73] are based on the smooth surfaces of baize, wooden plane transparent Perspex and rough sandpaper, whereas the ones from Lajeunesse et al.[74] are obtained from experiments with basal surfaces of sandpaper and glass. We can observe that generally the numerical results are well corroborated by the experimental observation. For the run-out, the numerical results are closer to the results from Lube et al.,[73] while for the final height, the numerical results are between those from Lube et al.[73] and Lajeunesse et al.[74] Note that the boundary particle method is essentially the case of $\mu = \tan(0.79\phi)$ in generalized frictional boundary.[56] The simulation results appear somehow inconsistent due to the fact that two different values of μ can yield similar results. This phenomenon will be explained in the next section through parametric analyses varying μ.

7.1.2 | The influence of the surface friction

With the generalized frictional boundary, it is possible to investigate the influence of the boundary friction on the flow dynamics and the final deposit shape. Here we carry out five groups of simulations with varying surface friction coefficient, that is, $\mu = 0.9$, 0.754, 0.5, 0.2, and 0.1. In each group, nine simulations with different aspects ratios are performed.

Lube et al.[73] noted that the collapse process and the final shape are generally independent of the surface over which the granular material flows, which indicates that the flow is independent of the surface frictional. This observation is obtained from the experiments but is against our common intuition. By changing the surface friction in our simulation, we can give a clear explanation. The normalized run-out distance and final height from the simulations with different aspect ratio and surface friction are shown in Figure 8. It is found that when the surface friction μ is between 0.9 and 0.5, the run-out and final height are almost independent of the surface friction. This range of surface friction corresponds to a frictional angle range between 42.0° and 26.6°, which is the common surface friction angle for many materials. Consequently, in the experiments, Lube et al. observed that the flow process and final deposit are independent from the surface friction.[73] Similar observation can be found in Sheikh et al.[46] and the conclusion of effective frictional angle conducted from del Castillo et al. [56] is naturally included here.

However, once the surface friction becomes even smaller, it starts to have greater impact on the collapse process. It can be observed from Figure 8 that the run-out distances from the simulations with $\mu = 0.1$ and 0.2 are significantly larger than the run-out distances from the simulations with larger surface friction coefficients. Consistently, the final heights are much smaller. It is also noted that the deposition patterns are different for collapses over surfaces with high and low friction. The 3D simulation of a collapsing column with $a = 1$ for $\mu = 0.754$ and $\mu = 0.2$ at time $t = 0.5$ s is shown in Figure 9, where h and r means the current height and radius of the model, respectively. In Figure 9, the deposition patterns of the two collapses both have cone shapes. Nevertheless, the moving of the particles with $\mu = 0.754$ is almost stopped at time $t = 0.5$ s, while the motion of the particles with $\mu = 0.2$ continues. In addition, at the outer edge of the deposit, a small number of scattered particles can be observed, which are located in the spreading regime of the deposit. To better illustrate the spreading regime, the cross-sectional views of the deposits are shown in Figure 10. It can be seen that there are three different regimes, including the undisturbed regime, the spreading regime and their intermediate regime. The undisturbed regime will be flattened with the decreasing μ values, and the spreading regime is a thin thickness and formed

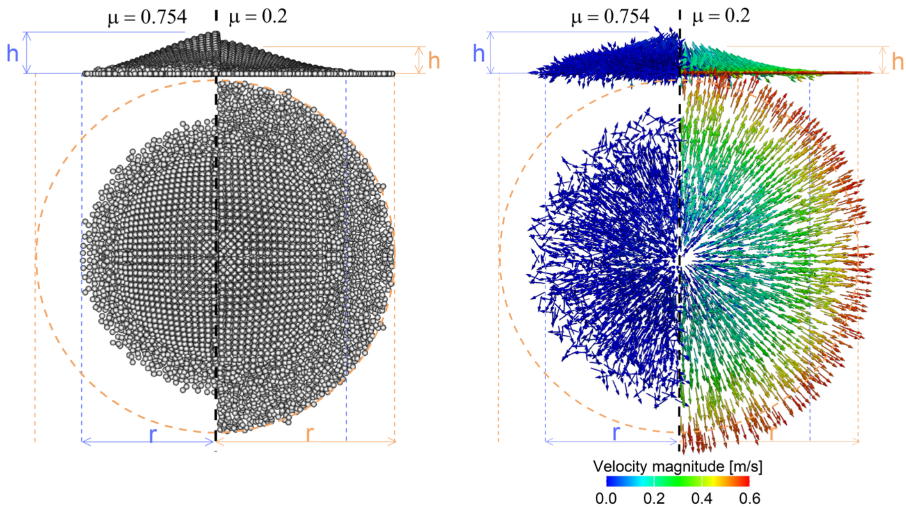

FIGURE 9 3D granular column with $a = 1$ at time $t = 0.5$ s: left half with $\mu = 0.754$ and right half with $\mu = 0.2$.

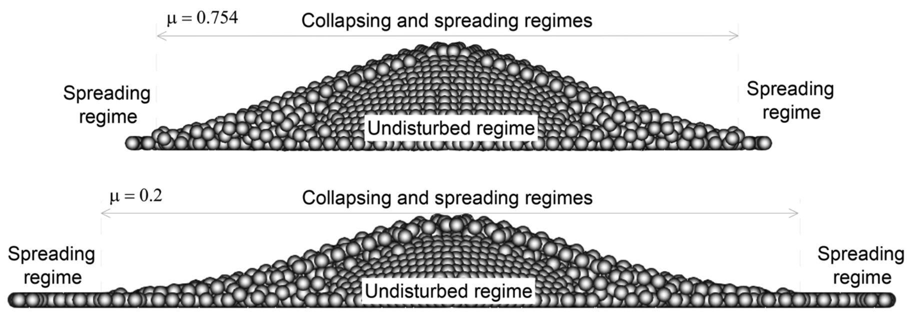

FIGURE 10 Cross-sectional views of the deposit for 3D granular column with $a = 1$ at time $t = 0.5$ s.

by spreading particles. The comprehensive discussions on these different regimes can be found in Sheikh et al.[46] Clearly, it is found that in the simulation with lesser surface friction, the radius of the spreading regime will be more significant, as shown in Figure 8. When the frictional coefficient is too small, all the materials will flow in the collapse process, as the resistance from the boundary is insufficient. In addition, more attempts related to frictional boundary methods can be found in Refs. 44–46.

7.1.3 | Remarks

Although the granular collapse is a well-studied numerical case, through this case, the following findings are drawn: (1) the proposed generalized frictional boundary is validated and gives results well corroborated by experiments; (2) it reveals why in the experiment the type of surface does not change the collapse process and the final deposit because only when the surface friction is smaller than a threshold, then it starts to have significant influence on the granular flow.

7.2 | Failure of cohesive soil

For cohesive soils, SPH simulations usually employ the artificial stress method to mitigate tensile instability. In this section, the capability of the regularized SPH model for problems with cohesive soils is demonstrated. The numerical case is taken from Bui et al.,[10] where a cohesive soil block undergoes large deformation due to self-gravity, as shown in Figure 11. The initial height and width of the soil are 2 and 4 m, respectively. The particle spacing is $\Delta x = 0.04$ m. The material parameters are the same as those in Bui et al.[10]: $E = 1.8$ MPa, $\nu = 0.33$, friction angle $\phi = 25°$, cohesion $c = 5$ kPa, $\psi = 0°$, and $\rho = 1850$ kg/m^3. The free-slip and fixed boundary methods are adopted to modeling the left wall and the

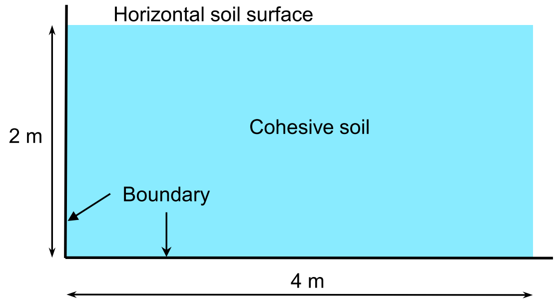

FIGURE 11 The schematic diagram of the cohesive soil model.

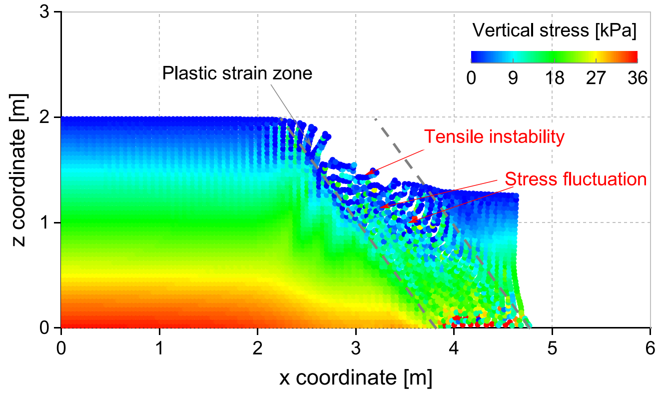

FIGURE 12 Vertical stresses at final stage of collapse investigated by SPH schemes.

bottom plane in this case, respectively.[23] In addition, the damping method is employed to obtain correct initial in situ stresses.[78]

The simulation results obtained by SPH scheme without any special treatment is given in Figure 12. The tensile instability can be clearly observed from the results, nonphysical crackings are found in the tensile regions. Another observation is that the stress fluctuations can be found in the area with large plastic deformation, that is, the area between the two dashed lines in Figure 12. This indicates that the original SPH suffers from tensile instability and low stress accuracy in the plastic region; thus, enhancements are required. In previous studies, several contributions have been made to present good predictions on the behaviors of cohesive soil.[10,70,72] Here, only PST approach is focused on.

7.2.1 | Parameters analysis for PST

As mentioned in Section 5.1, it is necessary to shift the particles with different shifting values depending on their positions. Following the contributions in Refs. 41, 42, 65, Equation (25) is employed to mitigate the tensile instability in this paper. Since the structure of Equation (25) has been tested by various studies,[41,42,65] this section only focuses on the parameters analysis of Equation (25) for solid computations.

To show the necessity of Equation (25), seven SPH simulations are carried out: (1) particles are shifted by PST scheme in Equation (25), and (2-7) all particles will be shifted by the same shifting vectors by using the following equation as:

$$\nabla \bar{C}_i = \sum_j \frac{m_j}{\rho_j} \left[A + B \left(\frac{W_{ij}}{W_{ij}(\Delta \mathbf{x})} \right)^n \right] \nabla_i W_{ij}, \quad \text{if } \|\mathbf{x}_i - \mathbf{x}_j\| < 2h \tag{34}$$

where A and B are constants. In Equation (34), the first term represents the concentration gradient value, and the second term is used to avoid the extreme particles clumping where the concentration gradient value begins to decrease to zero.[41] The simulation results obtained by Equation (34) with the given parameters in Table 1 are named as "PST1-PST6,"

TABLE 1 Parameters analysis for Equation (34).

$A = 0.0, B = 0.01$	PST1.	$A = 0.0, B = 0.1$	PST2.	$A = 0.0, B = 0.2$	PST3.
$A = 1.0, B = 0.0$	PST4.	$A = 0.5, B = 0.2$	PST5.	$A = 1.0, B = 0.2$	PST6.

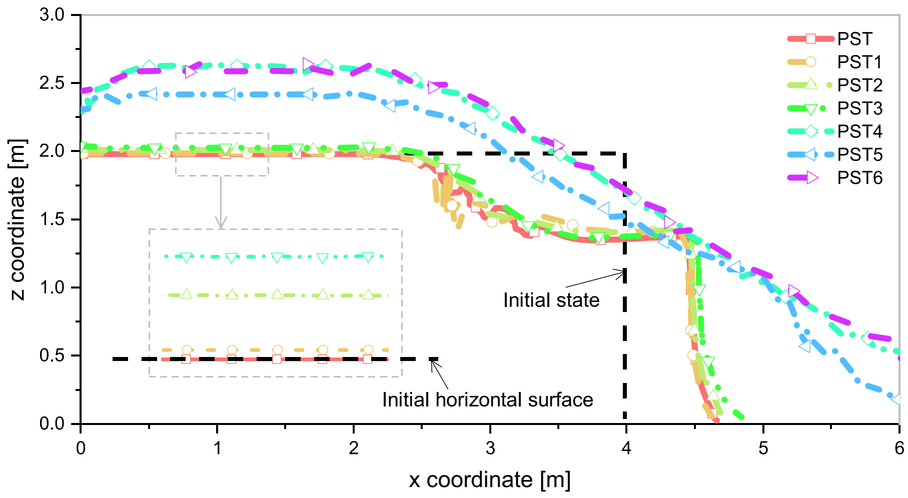

FIGURE 13 The free-surface profiles at final stage of collapse investigated by SPH schemes. SPH, smoothed particle hydrodynamics.

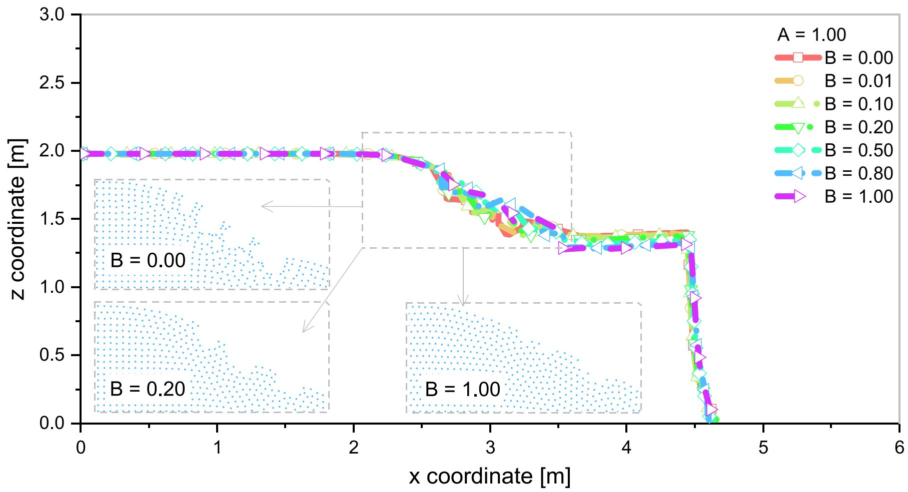

FIGURE 14 The free-surface profiles at final stage of collapse investigated by SPH schemes. SPH, smoothed particle hydrodynamics.

respectively. All free-surface profiles at the final deposit are plotted in Figure 13. In the simulations, unrealistic dilatancy (cases obtained by schemes of PST1-PST6) will be observed in the upper-left corner of the model, where the free surface is located in the region with almost no deformation. In addition, similar surface profiles between the cases obtained by PST and PST1 can be found, which suggests that the free-surface particles should be shifted by a minimal shifting value. However, for the scheme of PST1, strong tensile instability can be observed at the plastic strain zone, which is similar to that in Figure 12. Thus, for PST scheme, it is necessary to give special treatments to the free surface and its vicinity particles in solid computations.

Meanwhile, A and B can also represent the coefficients corresponding to contribution gradient and f_{ij} in Equation (25), respectively. In Equation (25), the original f_{ij} employed by Lind[41] is used to avoid the numerical instability caused by $\nabla \bar{C}_i \approx 0$. To test the effects of f_{ij} in solid computations, a group of tests is investigated by the scheme of Equation (25) with $A = 1.00$ and $B = 0.00 - 1.00$. The free-surface profiles at the final deposit state are shown in Figure 14. It can be found that similar free-surface profiles can be obtained. However, for the case of $B = 0.00$, the tensile instability can still be observed in the region of the free surface. With the increased B values, tensile instability on the free surface is mitigated

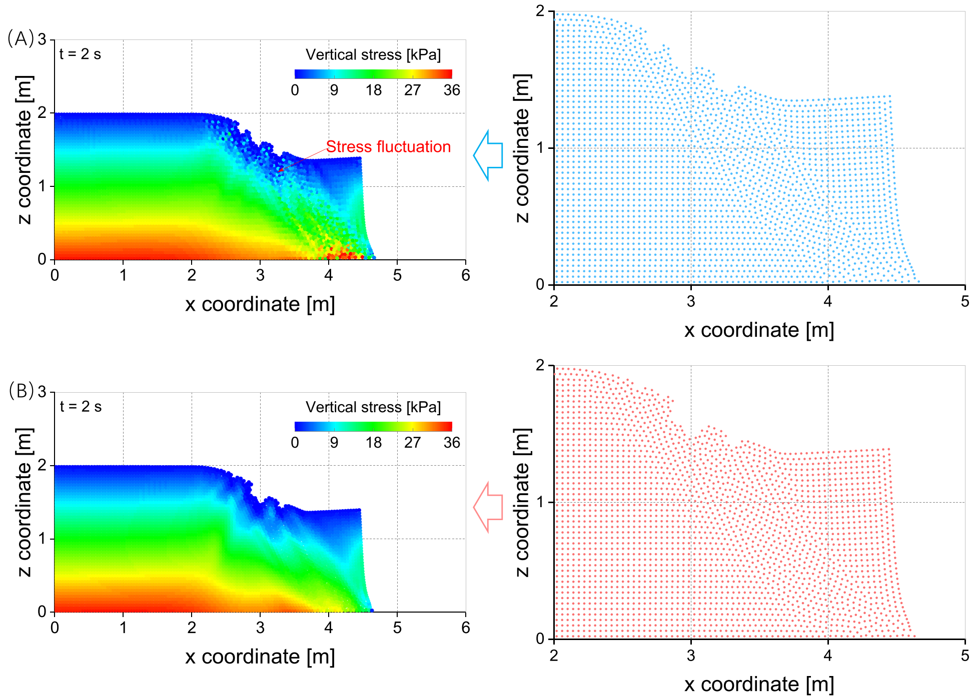

FIGURE 15 Vertical stresses and kinematic profiles at final stage of collapse investigated by proposed SPH schemes. (A) with PST; (B) with PST + SRT. PST, particle shifting technique; SPH, smoothed particle hydrodynamics; SRT, stress regularization technique.

progressively. Considering that a larger value of B is equivalent to adding a larger value of repulsive force in the calculation, a suitable B value should be carefully decided. Since there is no significant difference between the concerned simulation results obtained by schemes with $B = 0.1 - 0.5$, $B = 0.2$ is selected in this paper.

7.2.2 | Performance of the proposed regularization techniques

To test the proposed SPH scheme, two simulations are carried out: (1) only with the PST method and (2) PST coupled with the stress regularization technique (SRT). In Figure 15, the stress distributions obtained by the aforementioned schemes are shown on the left side of the figure, respectively. The zoom-in snapshots of particle distribution in the plastic strain zone are shown on the right side. It is found that with particle shifting, a uniform particle distribution is maintained. However, the stress results still have some fluctuations. This indicates that although PST can sufficiently suppress tensile instability, it is unsuitable for soil–structure interaction problems if accurate stress results are desired. Finally, if particle shifting is combined with stress regularization, the simulation is free from tensile instability and gives smoother stress results. Notable, the scheme of PST coupled with SRT does not completely resolve the issue of nonphysical displacement of particles on the free surface in the tensile region, which can be improved in the future.

The comparison of the stress results simulated by two SPH schemes (PST and PST coupled with SRT, respectively) is given in Figure 16, where the vertical stresses along the two vertical lines, that is, $x = 3$ m and $x = 4$ m, are plotted. After the collapse, there is no analytical solution for the stress distribution, as it is not geostatic. Thus, the geostatic stress results of the two final deposits, calculated by $\rho g z$, are only defined as the reference results here and are shown in the figures in the form of lines, where z is distance from the soil from the soil top-surface at the same x location. The symbols stand for the data collected directly from the simulation results. It is found that the stress results from the simulation with PST have slight fluctuations. Correspondingly, the scheme of PST coupled with SRT bring uniform and smooth stress distribution. The stress data of the two groups are not coincident with the reference results, it can be explained that the soil structure undergoes the remodeling and the soil is no longer in a geostatic status.

From Figure 13 to 16, for the simulation of cohesive soil, we find that the PST method can provide good results of kinematic profiles but leaves some stress fluctuations. Combining with the existing methods, it is shown that PST coupled with SRT can provide nice and smooth stress field. Though this scheme does not completely resolve the issue of nonphysical

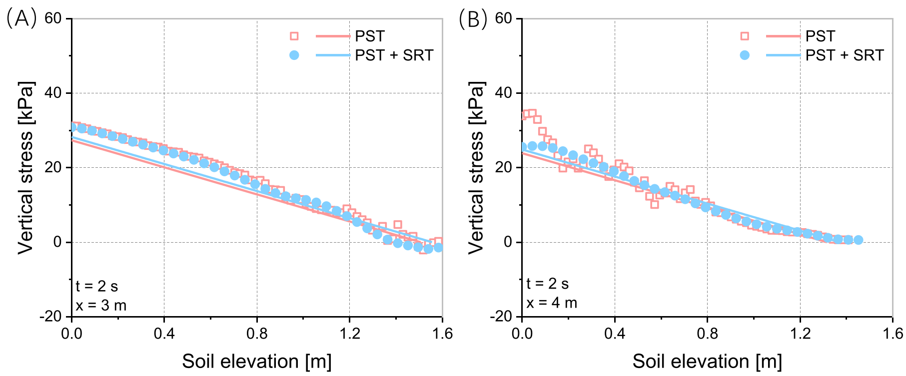

FIGURE 16 Vertical stresses at final stage of collapse investigated by proposed SPH schemes. (A) Horizontal coordinate $x = 3$ m; (B) horizontal coordinate $x = 4$ m.

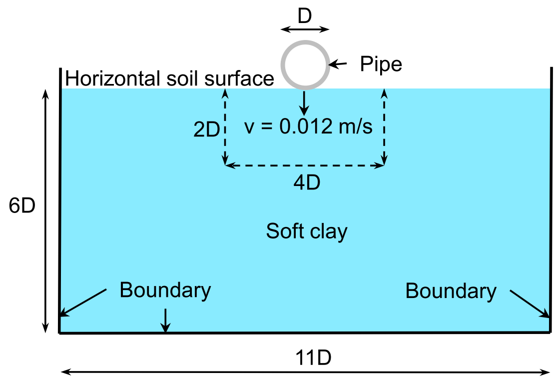

FIGURE 17 The schematic diagram of the pipe vertical penetrates into soft clay.

displacement of particles on the free surface, the simulation results show promising potential especially to handle the soil–structure interaction force.

7.3 | Pipe penetration into soft clay

Interactions between pipes and seabeds are commonly found in offshore engineering. Here a problem of pipe vertically penetrating into soft clay is simulated with the presented SPH method to check its performance in model large-deformation soil–structure interaction. The experiment was carried out by Dingle et al.,[75] and was later modeled using the Arbitrary Lagrangian Eulerian method.[3] As shown in Figure 17, the pipe is considered as a rigid structure and penetrates down into the soil. The height and width of the soil model are 6 and 11 times the diameter of pipe $D = 0.8$ m, respectively. The soil in this case is soft clay under undrained condition, typical for seabed; therefore, the friction angle is zero. The material parameters are given following[3,75] the buoyant weight of the clay is $6.5\ \mathrm{kN/m^3}$; the initial cohesion of the soil is taken as $s_{u0} = 2.3 + 3.6z$ kPa, where z is the soil depth; Young's model $E = 500s_{u0}$; and Poisson's ratio $\nu = 0.499$.

The shear strength at the pipe–soil interface is s_u, which is the remolded shear strength to account for the strain rate effect and strain softening. Following Chatterjee et al.,[3] the following relation is employed to determine the shear strength

$$s_u = \left[1 + \mu_r \log\left(\frac{\max(\dot{\gamma}_{max}, \dot{\gamma}_{ref})}{\dot{\gamma}_{ref}}\right)\right] \times \left[\frac{1}{S_t} + \left(1 - \frac{1}{S_t}\right)\exp\left(\frac{-3k}{k_{95}}\right)\right] s_{u0} \tag{35}$$

where the values of the parameters are the same as those in the test[3]: rate parameter $\mu_r = 0.1$; $\dot{\gamma}_{\max} = \dot{\varepsilon}_1 - \dot{\varepsilon}_3$, where $\dot{\varepsilon}_1$ and $\dot{\varepsilon}_3$ are the major and minor principal strain rates; $\dot{\gamma}_{ref} = 3 \times 10^{-6}$ is the reference strain rate; the soil sensitivity $S_t = 3.2$ is defined as the inverse of reduction coefficient μ_c; $k_{95} = 10$ represents the cumulative shear strain for 95% shear strength degradation. In addition, k is the accumulated absolute plastic shear strain, its equation can be found in Peng et al.[37]

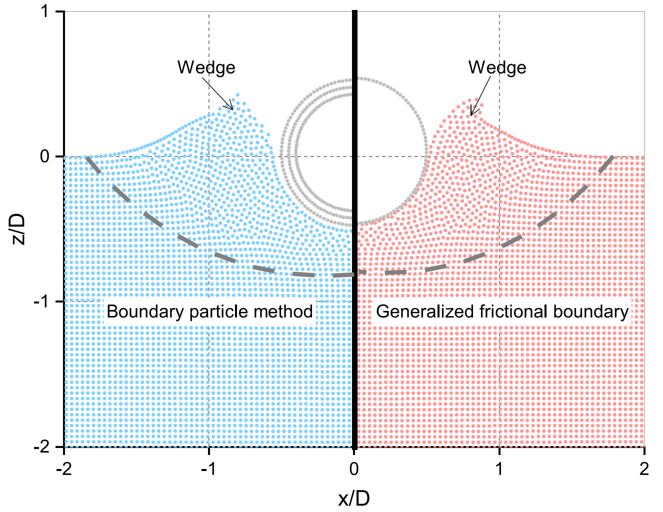

FIGURE 18 Position profiles of the deposits investigated by SPH schemes. SPH, smoothed particle hydrodynamics.

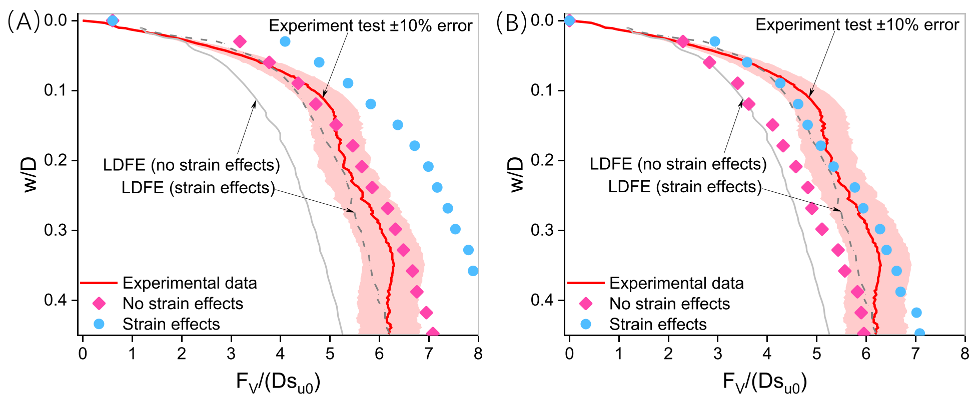

FIGURE 19 Vertical resistance obtained from SPH modeling and experimental studies.[3,75] (A) Proposed SPH scheme with the boundary particle method; (B) proposed SPH scheme with the generalized frictional boundary. SPH, smoothed particle hydrodynamics.

For the SPH modeling, the interaction between soil and structure is modeled using the boundary particle method and the generalized frictional boundary, respectively. Four additional simulations are carried out to check the influence of strain effects (including rate effect and strain softening). Since the friction angle is equal to zero, the maximum shear strength s_u along the pipe–structure interface in Equation (35) is equal to the cohesion c, and the reduction coefficient is $\mu_c = 1/S_t$[3]; the penetration coefficient $\xi = 0.001$ is adopted.[44]

Figure 18 shows the kinematic profiles in regions close to the pipe–structure interface at selected time instances. The results obtained by the regularized SPH scheme with two boundary methods are shown in the left half and right half of the figure, respectively. The dashed lines are the boundaries between the disturbed and undisturbed regions.[3] It can be found that the two wedge-shaped soil deposits on the free surfaces are formed, which are similar to the deformation pattern in the experimental and previous numerical tests.[3,75] Meanwhile, the kinematic profiles obtained by the two SPH approaches are similar. Both methods work well, as no tensile instability can be observed. This indicates that the particle shifting method can be a feasible alternative to the conventional artificial stress method.

In addition, the reaction force in the vertical direction obtained using the nonslip boundary method can be given as

$$F_V = -\mathbf{n}_V \cdot \sum_b m_b \sum_i m_i \left(\frac{\sigma_b}{\rho_b^2} + \frac{\sigma_i}{\rho_i^2} + \Pi_{bi}\mathbf{I} \right) \cdot \nabla_b W_{bi} \tag{36}$$

where the subscript i represents the material particles in the support domain of a boundary particle b; m_b is the mass of the boundary particle; $\mathbf{n}_V$ is the unit vector in the normal direction of the pipe surface. On the other hand, the vertical

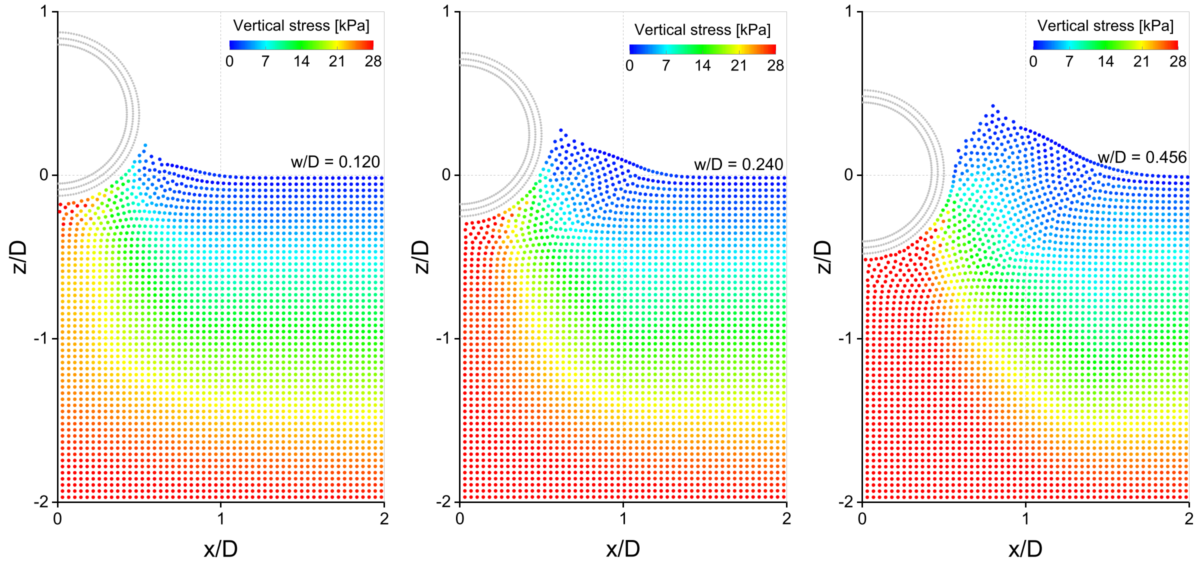

FIGURE 20 The vertical stresses and position profiles of the penetration process (with nonslip boundary).

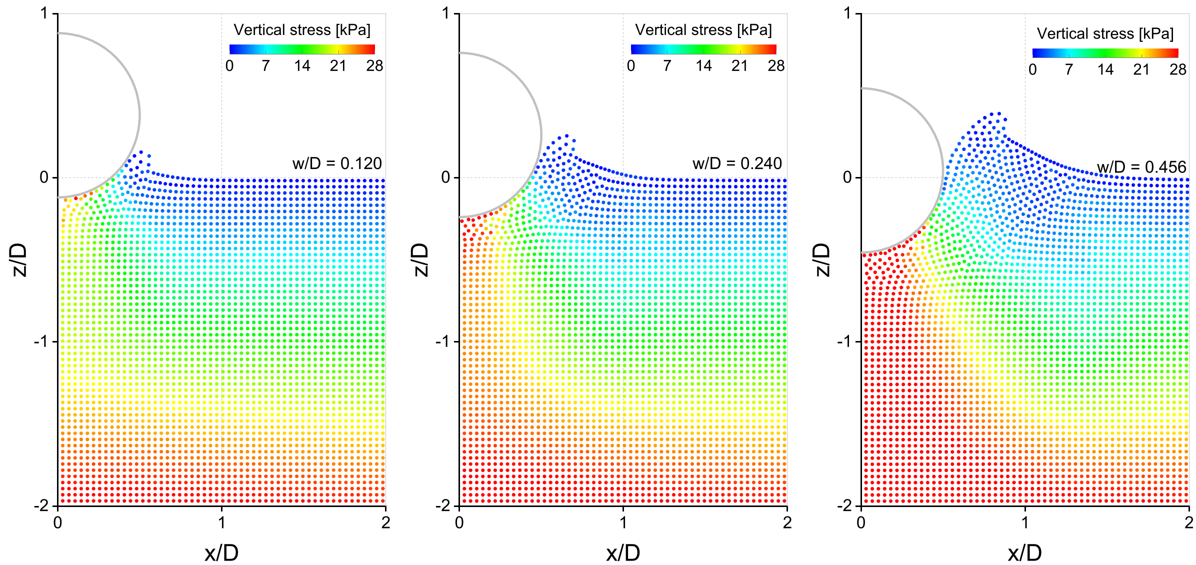

FIGURE 21 The vertical stresses and position profiles of the penetration process (with contact boundary).

reaction force obtained using the generalized frictional boundary can be calculated directly as

$$F_V = \mathbf{n}_V \cdot \sum_i m_i \mathbf{a}_i^e \tag{37}$$

Moreover, there are several ways to calculate the normal force, as shown in Refs. 45 and 46.

The normalized vertical reaction force $F_V/(Ds_{u0})$ and the normalized embedding distance w/D are plotted in Figure 19 for both the SPH simulations and the test, with w being the distance of pipe penetration and s_{u0} being the shear strength of the soil at the pipe invert. In the figure, the experimental data are from Dingle et al.[75]; LDFE stands for the simulation results obtained by Arbitrary Lagrangian Eulerian method.[3] From Figure 19, it can be seen that in the simulations, the resistance results from the SPH scheme of the boundary particle method are slightly overestimated. However, the simulation results obtained from the generalized frictional boundary are closer to the experimental data, which can simulate soil–structure interaction with good accuracy. Meanwhile, in all the simulations, the reaction forces without strain effects are lower than those with strain effects. This observed trend is also well corroborated by the results from Chatterjee et al.[3]

The main advantage of the SPH simulation is to conveniently capture the large deformation when the pipe is pushed into soil. The evolution process of vertical stress and position profile for soil are illustrated in Figures 20 and 21, both are simulated using the regularized SPH scheme but with different interaction treatment methods. With the same penetration depth, for instance $w/D = 0.118$, it is clear to find that the vertical stress of soil directly below the pipe in Figure 20 is overall larger than that in Figure 21.

This phenomenon shows that, to model soil–structure interaction, it is necessary to consider the actual contact condition along the interface.

8 | CONCLUSIONS

This paper presents a regularized SPH method for large-deformation soil–structure interaction. For the problem related to tensile instability and stress noise, the combination of particle shifting and stress regularization is proposed to maintain uniform stress and particle distribution. A generalized frictional boundary is proposed based on a hybrid contact method, which can model the frictional-cohesive contact along soil–structure interfaces accurately. The particle shifting method is enhanced to be consistent with the generalized frictional method.

Several numerical examples are presented to check the performance of the regularized SPH scheme. Numerical results are compared with experimental and numerical results from literature. The following conclusions are drawn. (1) Particle shifting is a feasible alternative to the artificial stress method in preventing tensile instability, with the additional benefits of less stress fluctuation and more uniform particle distribution; (2) particle shifting combined with C^1 stress regularization can give good results for large deformation analysis of cohesive and cohesionless geomaterials with smooth stress results free of any instability; (3) the generalized frictional boundary can accurately model cohesive-frictional contact along the soil–structure interfaces with arbitrary contact paramerers; (4) the presented regularized SPH scheme with the generalized frictional boundary gives rise to reliable simulations of large deformation soil–structure interaction problems.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author upon reasonable request.

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